Construction and Interpretation of a Hertzsprung–Russell Diagram Using Gaia DR3 Stellar Data

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Abstract

The relationship between stellar temperature and intrinsic brightness is central to stellar astrophysics. In this study, stellar data from the Gaia mission were analyzed to construct a Hertzsprung–Russell (HR) diagram using Python-based computational tools. A dataset containing stellar color index, apparent magnitude, and parallax measurements was obtained from the Gaia Archive. Stellar distances were calculated from parallax values, and apparent magnitudes were converted into absolute magnitudes in order to estimate intrinsic stellar brightness. These parameters were then used to construct an HR diagram, revealing major stellar populations such as the main sequence and red giant stars. The project demonstrates how modern astronomical datasets can be explored through computational analysis and provides an introductory but research-oriented exercise in observational stellar astronomy.

1. Introduction

Stars are the fundamental building blocks of galaxies and serve as natural laboratories for studying physical processes under extreme conditions. Their observable properties—such as color, brightness, and distance—allow astronomers to infer physical quantities including temperature, luminosity, and evolutionary stage. One of the most useful tools for visualizing these relationships is the Hertzsprung–Russell (HR) diagram, which plots stellar brightness against temperature or color and reveals how stars are distributed across different stages of stellar evolution.

Most stars occupy a diagonal region of the HR diagram known as the main sequence, where hydrogen fusion occurs in the stellar core. More evolved stars, such as red giants, occupy different parts of the diagram because their radii, temperatures, and luminosities differ significantly from those of main-sequence stars. As a result, the HR diagram is not only a classification tool but also a compact visual summary of stellar evolution.

The Gaia mission has made it possible to examine stellar populations on an unprecedented scale by providing precise astrometric and photometric measurements for more than a billion stars in the Milky Way. In this project, a subset of Gaia DR3 stellar data was analyzed using Python to construct and interpret an HR diagram. The project combines observational data, basic

astrophysical concepts, and scientific computing methods to investigate the relationship between stellar color and intrinsic brightness.

The study of stellar populations is essential for understanding both stellar evolution and the large-scale structure of galaxies. The HR diagram remains one of the most informative and visually powerful tools in astrophysics because it connects observable quantities—color and brightness—to underlying stellar physics. Constructing an HR diagram from real survey data therefore provides more than a coding exercise; it offers a direct entry point into research methodology in astronomy.

The Gaia mission has transformed modern stellar astronomy by making precise astrometric and photometric data widely accessible. Even a small subset of Gaia DR3 measurements can reveal fundamental astrophysical structure when analyzed carefully. This project reflects that idea by combining observational data, physical interpretation, and computation. It demonstrates how accessible scientific tools can be used to engage meaningfully with real astrophysical datasets and motivates further study in stellar and galactic astronomy.

2. Literature Review

The Hertzsprung–Russell (HR) diagram is one of the most fundamental tools in stellar astrophysics for understanding the relationship between stellar luminosity and temperature. The diagram was originally developed by Ejnar Hertzsprung and Henry Norris Russell in the early twentieth century and has since become essential for studying stellar evolution and stellar classification. The HR diagram reveals that most stars lie along a diagonal band known as the main sequence, where stars generate energy through hydrogen fusion in their cores. More evolved stars, such as red giants and supergiants, occupy different regions of the diagram due to their larger radii and higher luminosities. Compact stellar remnants known as white dwarfs appear in the lower part of the diagram because they are hot but intrinsically faint. Detailed discussions of the HR diagram and stellar evolution can be found in standard astrophysics texts such as Carroll and Ostlie (2017).

Early HR diagrams were constructed using relatively small samples of nearby stars with measured distances. The development of large astronomical surveys has significantly expanded the available data for stellar population studies. Modern surveys provide accurate measurements of stellar brightness, color, and distance, allowing astronomers to construct detailed color–magnitude diagrams for large numbers of stars.

One of the most important contributions to modern stellar astronomy is the Gaia mission conducted by the European Space Agency. The Gaia mission provides precise astrometric and photometric measurements for more than a billion stars in the Milky Way galaxy. Gaia Data Release 2 and Data Release 3 have enabled astronomers to construct high-precision observational HR diagrams and study the structure and stellar populations of the Milky Way with unprecedented detail (Gaia Collaboration, 2018; Gaia Collaboration, 2022). These datasets have become essential resources for modern research in stellar astrophysics.

Recent studies have also emphasized the importance of computational methods for analyzing large astronomical datasets. Programming environments such as Python, along with scientific libraries including Pandas, NumPy, and Matplotlib, have become widely used tools for astronomical data analysis and visualization. These tools allow researchers to efficiently

process large stellar catalogs and generate HR diagrams that reveal important patterns in stellar populations. Educational resources such as OpenStax Astronomy (2016) highlight the role of HR diagrams in understanding stellar properties and demonstrate how observational data can be used to explore fundamental relationships between stellar temperature and luminosity.

In this study, Gaia DR3 data are used to construct and analyze a Hertzsprung–Russell diagram for a selected stellar sample. By combining observational data from the Gaia Archive with computational analysis using Python, the study demonstrates how large astronomical datasets can be used to investigate stellar populations and illustrate key concepts in stellar astrophysics.

Large observational HR diagrams constructed from Gaia data have revealed detailed stellar structures including the main sequence, red giant branch, and white dwarf sequences, enabling new insights into the structure and evolution of the Milky Way galaxy.

2. Data Source

The dataset used in this project was obtained from the Gaia Archive, the public data portal for the Gaia mission. Gaia provides high-precision measurements of stellar positions, parallaxes, and photometric properties. For this analysis, a subset of stars was selected using the following parameters from the Gaia DR3 catalog: BP−RP color index, G-band apparent magnitude, and parallax.

The final dataset used in this analysis contains approximately 30,000 stars selected from the Gaia DR3 archive.

The three parameters used are listed below:

Parameter	**Description**
bp_rp	Stellar color index used as a proxy for stellar temperature
phot_g_mean_mag	Apparent brightness measured by Gaia in the G band
parallax	Astrometric measurement used to calculate stellar distance

Table 1. Parameters selected from the Gaia DR3 dataset for constructing the HR diagram.

3. Stellar Parameters and Physical Background

The physical interpretation of the HR diagram depends on three core quantities: stellar brightness, distance, and temperature. In the Gaia dataset, these quantities are represented indirectly through apparent magnitude, parallax, and color index.

Apparent magnitude describes how bright a star appears from Earth. However, apparent brightness depends strongly on distance. A distant star may appear faint even if it is intrinsically very luminous. To compare the true brightness of different stars, astronomers use absolute magnitude, defined as the magnitude a star would have if placed at a standard distance of ten parsecs.

The relationship between apparent magnitude and absolute magnitude is given by the distance modulus equation: $M = m - 5 \log10(d) + 5$ where M is the absolute magnitude, m is the apparent magnitude, and d is the distance in parsecs. In astronomy, the magnitude scale is inverse: smaller magnitude values correspond to brighter objects, while larger values correspond to fainter objects. For this reason, HR diagrams are commonly plotted with the magnitude axis inverted so that brighter stars appear toward the top of the figure.

Distance was determined using parallax. Parallax is the apparent shift in the position of a nearby star against distant background stars as Earth moves around the Sun. The stellar distance in parsecs is given by d = 1/p when parallax is measured in arcseconds. Because Gaia provides parallax in milliarcseconds, the distance becomes $d = 1000/p$.

Stellar temperature is represented in this project by the BP−RP color index. Lower BP−RP values correspond to hotter, bluer stars, while larger BP−RP values correspond to cooler, redder stars. Thus, the color index provides an observational proxy for stellar temperature without requiring direct spectroscopic measurements.

4. Methodology

The analysis was performed using Python with the libraries Pandas, NumPy, and Matplotlib. Pandas was used to read the CSV file exported from the Gaia Archive and to filter invalid measurements. NumPy was used to calculate distances and absolute magnitudes, while Matplotlib was used to visualize the processed data.

The workflow followed in this project was as follows:

1. A subset of Gaia DR3 stellar data was downloaded from the Gaia Archive.
2. The dataset was imported into Python using Pandas.
3. Stars with invalid or unreliable values were removed by filtering on parallax and by dropping missing entries.
4. Distances were computed from parallax values.
5. Apparent magnitudes were converted into absolute magnitudes.
6. A scatter plot of BP−RP color index against absolute magnitude was generated to construct the HR diagram.
7. The vertical axis was inverted so that brighter stars appear toward the top of the graph, matching standard astronomical convention.

In addition to the final plot, marker size, transparency, and color mapping were adjusted to improve the visual separation of stellar populations. These choices made the diagram easier to interpret and more consistent with the appearance of research-style HR diagrams.

5. Results and Interpretation

The final HR diagram clearly reveals the non-random structure of stellar populations in the Gaia sample. The most prominent feature is the main sequence, which appears as a diagonal band extending from hotter and intrinsically brighter stars toward cooler and intrinsically fainter stars. This region represents stars that are steadily generating energy through hydrogen fusion in their cores.

A second notable population appears above the main sequence. These stars are cooler in temperature but still very bright, which is characteristic of red giant stars. Their position on the

diagram reflects the fact that luminosity depends not only on temperature but also strongly on stellar radius. A star can therefore be relatively cool and still highly luminous if it has expanded to a very large radius.

The lower region of the plot contains stars with low intrinsic brightness. These are primarily faint, low-mass stars, along with additional scatter caused by the limited sample size and by measurement uncertainties. Even with a relatively small dataset, the HR diagram successfully captures the major stellar trends expected from theory and from large stellar surveys.

Figure 1 presents the final HR diagram constructed in this project. The x-axis shows the BP−RP color index, while the y-axis shows absolute magnitude. Because the magnitude scale is inverted, brighter stars appear higher in the diagram. The color mapping applied to the points further emphasizes the gradual shift from hotter blue stars to cooler red stars.

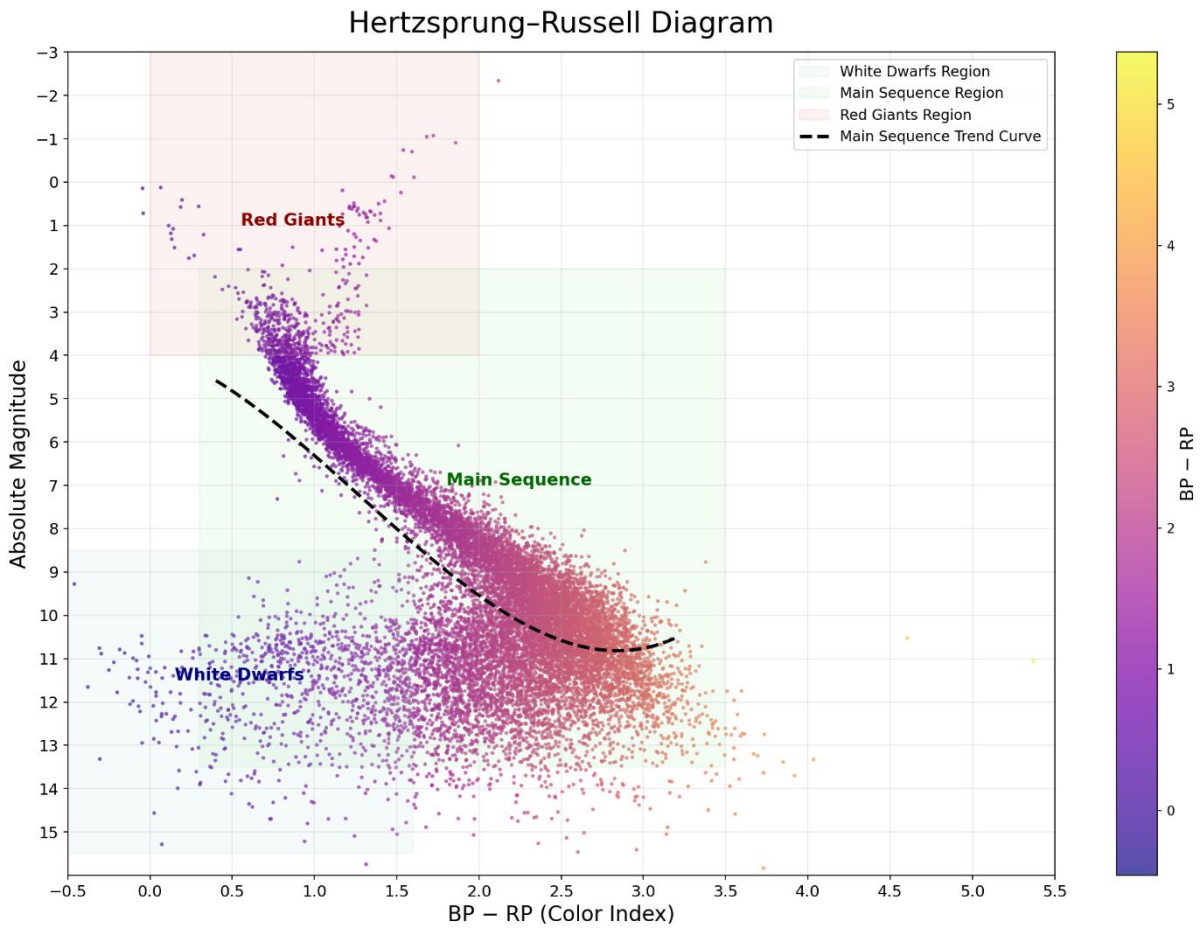

Figure 1. Hertzsprung–Russell diagram constructed from Gaia DR3 stellar data, showing BP−RP color index versus absolute magnitude.

6. Limitations of the Dataset

Although the HR diagram constructed in this project clearly illustrates major stellar populations, several limitations should be acknowledged. First, the analysis used only a small subset of the full Gaia DR3 archive. A larger sample would provide a denser and more complete stellar distribution, making features such as subgiants or white dwarfs easier to identify.

Second, measurement uncertainties in parallax values affect the calculated distances and therefore propagate into the absolute magnitudes. In professional analyses, more rigorous quality filters—such as parallax signal-to-noise ratio or photometric uncertainty constraints—are often applied to reduce these effects. In this project, only simple filtering was used in order to keep the workflow clear and accessible.

Finally, BP−RP was used as a temperature proxy instead of direct stellar effective temperature measurements. While this is a standard and useful observational approach, it remains an approximation. Despite these limitations, the final diagram successfully demonstrates the expected relationship between stellar color and intrinsic brightness.

7. Future Work

Several extensions could make this study more comprehensive. A larger Gaia sample could be used to produce a more detailed and statistically robust HR diagram. Applying stricter quality cuts would improve the reliability of the stellar positions in the diagram. Additional Gaia parameters or external spectroscopic data could also be incorporated to explore stellar metallicity, age, or kinematics.

Another natural extension would be to isolate specific stellar populations such as white dwarfs, red clump stars, or stars belonging to open clusters. More advanced computational approaches, including clustering methods or machine-learning based classification, could be applied to identify stellar groups automatically. These extensions would shift the project from an introductory data-analysis exercise toward a more fully developed observational astrophysics study.

8. Conclusion

This project demonstrates how real astronomical survey data can be used to investigate stellar properties through computational analysis. By combining Gaia DR3 data with Python-based processing, an HR diagram was constructed to visualize the relationship between stellar color and intrinsic brightness. The resulting diagram clearly shows the main sequence and a population of red giant stars, confirming the usefulness of the HR diagram as a tool for understanding stellar evolution.

Beyond the final graph itself, the project provided practical experience in accessing astronomical databases, cleaning observational data, performing astrophysically meaningful calculations, and interpreting stellar populations. As an introductory but research-oriented study, it provides a strong foundation for further work in observational astrophysics and scientific data analysis.

The study demonstrates how publicly available astronomical survey data can be used to perform meaningful astrophysical analysis using computational tools.

Appendix A. Core Python Workflow

The following code illustrates the core computational workflow used to generate the final HR diagram from Gaia DR3 data.

```
import matplotlib.pyplot as plt
import pandas as pd
import numpy as np
```

```
# Load data
data = pd.read_csv(r'C:\Users\parul\Desktop\30000-1970-11f1-8d67-bc97e148b76b-O-result.csv')

# Clean data
data = data.dropna(subset=['bp_rp', 'phot_g_mean_mag', 'parallax'])
data = data[data['parallax'] > 1]

# Distance in parsec
data['distance'] = 1000 / data['parallax']

# Absolute magnitude
data['absolute_magnitude'] = data['phot_g_mean_mag'] - 5 * np.log10(data['distance']) + 5

# Variables
color = data['bp_rp']
magnitude = data['absolute_magnitude']

# ---------------------------
# Select approximate main-sequence stars for fitting
# ---------------------------
# You can adjust these limits if needed
ms_data = data[
    (data['bp_rp'] > 0.4) &
    (data['bp_rp'] < 3.2) &
    (data['absolute_magnitude'] > 2.5) &
    (data['parallax'] > 0)&
    (data['phot_g_mean_mag'] < 18)
]

x_ms = ms_data['bp_rp']
y_ms = ms_data['absolute_magnitude']
degree = 3
coeffs = np.polyfit(x_ms, y_ms, degree)
poly = np.poly1d(coeffs)

x_curve = np.linspace(x_ms.min(), x_ms.max(), 300)
y_curve = poly(x_curve)
fig, ax = plt.subplots(figsize=(14, 10), dpi=150)

# Scatter plot of stars
sc = ax.scatter(
    color, magnitude,
    c=color,
    cmap='plasma',
    s=8,
    alpha=0.7,
    edgecolors='none'
```

```
)

# Add transparent region shading
# White dwarfs region
ax.fill_between(
    [ -0.5, 1.6 ], 15.5, 8.5,
    color='lightblue',
    alpha=0.12,
    label='White Dwarfs Region'
)

# Main sequence region
ax.fill_between(
    [ 0.3, 3.5 ], 13.5, 2.0,
    color='lightgreen',
    alpha=0.10,
    label='Main Sequence Region'
)

# Red giants region
ax.fill_between(
    [ 0.0, 2.0 ], 4.0, -3.0,
    color='lightcoral',
    alpha=0.10,
    label='Red Giants Region'
)

# Plot fitted curve for main sequence
ax.plot(
    x_curve, y_curve,
    color='black',
    linewidth=2.5,
    linestyle='--',
    label='Main Sequence Trend Curve'
)

# Labels for regions
ax.text(0.15, 11.5, 'White Dwarfs', fontsize=13, color='navy', weight='bold')
ax.text(1.8, 7.0, 'Main Sequence', fontsize=13, color='darkgreen', weight='bold')
ax.text(0.55, 1.0, 'Red Giants', fontsize=13, color='darkred', weight='bold')

# Axis settings
ax.set_title('Hertzsprung–Russell Diagram', fontsize=22, pad=15)
ax.set_xlabel('BP − RP (Color Index)', fontsize=16)
ax.set_ylabel('Absolute Magnitude', fontsize=16)

ax.set_xlim(-0.5, 5.5)
ax.set_ylim(16, -3)   # inverted HR diagram
```

```
ax.set_xticks(np.arange(-0.5, 5.6, 0.5))
ax.set_yticks(np.arange(-3, 16, 1))

ax.tick_params(axis='both', labelsize=12)
ax.grid(True, alpha=0.25)

# Colorbar
cbar = plt.colorbar(sc, ax=ax)
cbar.set_label('BP − RP', fontsize=14)

# Legend
ax.legend(fontsize=11, loc='upper right')
plt.tight_layout()
plt.show()
```

References

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